



LAVAE-Net: A deep LSTM-attention VAE framework for detecting ECG anomalies

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ABSTRACT

Electrocardiogram (ECG) abnormality detection is crucial for the early diagnosis of cardiovascular diseases. Traditional supervised-learning methods, however, requires abundant labeled abnormal samples, which are clinically scarce and costly to annotate. To overcome this limitation, we propose LAVAE-Net—an unsupervised ECG anomaly-detection model that fuses a Variational Autoencoder (VAE) with a Long Short-Term Memory (LSTM) network enhanced by an attention mechanism. LAVAE-Net trains exclusively on normal ECG data and identifies anomalies by learning the intrinsic latent distribution of normal cardiac dynamics. Architecturally, the encoder employs an LSTM backbone to capture long-term temporal dependencies, while an attention module adaptively highlights diagnostically salient features. The VAE encoder then maps the sequential ECG signal into a low-dimensional latent space and reparameterizes it into Gaussian-distributed latent variables. A GRU-based decoder reconstructs the original sequence from these latent variables, enabling the model to quantify deviations from the learned normal manifold. Training is driven by a composite loss that balances reconstruction error with Kullback–Leibler divergence, ensuring that latent representations remain both faithful and regularized. Evaluated on the public ECG5000 dataset, LAVAE-Net attains an F1 score of 99.4% for anomaly detection and maintains high precision for both normal and abnormal instances. Consequently, by eliminating the need for labeled anomalies, the model markedly improves clinical practicality and generalization thus providing a robust foundation for intelligent real-time ECG monitoring systems.

1. Introduction

Electrocardiogram (ECG) signals are indispensable physiological indicators in clinical practice, providing critical information for the early diagnosis and prevention of cardiovascular diseases [1]. The rapid proliferation of wearable sensors and the maturation of telemedicine platforms have further amplified the demand for automated, high-precision ECG anomaly detection [2,3]. However, the effectiveness of traditional machine learning methods is often hindered by the scarcity of labeled ECG data, particularly for abnormal heart rhythms. Annotating ECG data requires substantial expert knowledge and is both time-consuming and costly, leading to limited availability of labeled abnormal samples. Moreover, ECG data exhibit temporal variability, inter-individual differences, and persistent noise contamination, posing formidable challenges to conventional machine learning techniques and necessitating more robust analytical frameworks.

Recent advancements in autoencoder models have demonstrated

remarkable potential in anomaly detection [4]. By learning low-dimensional representations of data, autoencoders can effectively capture the characteristic distribution of normal samples, thereby identifying anomalies that deviate from normal patterns [5]. Nevertheless, basic autoencoder architectures still exhibit limitations when handling time-series data, particularly in capturing critical temporal features [6]. Variational autoencoders (VAEs) employ probabilistic latent space modeling, enhancing generative capability and feature disentanglement. Compared to traditional autoencoders, VAEs demonstrate superior noise robustness, overfitting prevention, and controllable generation [7], establishing them as a mainstream approach for time-series analysis and anomaly detection [8]. Additionally, attention mechanisms embedded within Convolutional Neural Networks (CNNs) have proven advantageous in sequence learning and anomaly detection [9]. By dynamically highlighting the most informative regions of the input or latent space, attention enables the model to focus on diagnostically relevant features, improving detection performance [10].

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However, despite these advancements, the reliance on labeled data remains a key limitation in most existing methods. In clinical settings, where abnormal ECG signals are often rare and difficult to label, methods that require abundant labeled data are impractical. To overcome the aforementioned limitations, we propose LAVAE-Net, a novel deep learning architecture that combines a VAE encoder with a unidirectional LSTM enhanced by attention mechanisms. By endowing the model with an enhanced capacity for learning and an elevated sensitivity to temporal dynamics, LAVAE-Net markedly improves the detection of abnormal ECG patterns. A unidirectional attention mechanism shrinks the model while retaining focus on key beats, and a dynamic threshold adapts to patient variation. Training combines Huber loss with KL divergence, boosting accuracy and robustness.

The main contributions of this work are as follows:

- (1) We propose an attention-enhanced LSTM-VAE architecture for ECG anomaly detection, which effectively captures long-range temporal dependencies while learning a structured and continuous latent space from normal cardiac patterns.
- (2) A normal-only anomaly detection framework is developed, enabling effective detection of abnormal ECG signals without relying on labeled abnormal samples during training.
- (3) Extensive robustness evaluations are conducted under realistic noise conditions, demonstrating that the proposed method exhibits graceful performance degradation in noisy environments.
- (4) Latent space visualization and interpretability analyses are provided to offer intuitive evidence of how abnormal ECG patterns deviate from the learned normal data manifold.

2. Related work

ECG abnormality detection is an important research direction in the field of medical artificial intelligence, which primarily depends on feature extraction for classification and are difficult to adapt to diverse ECG patterns. Recently, deep learning methods represented by CNN, Recurrent Neural Network (RNN) and their variants have demonstrated superior performance in ECG classification and anomaly detection.

2.1. ECG detection based on machine learning

Recent years have witnessed a surge of interest in compact yet accurate arrhythmia-detection systems. The classification device developed by Al-Shammary et al. [11], which uses the Chi-square Distance and the PSO algorithm, provides a high-precision and lightweight solution for arrhythmia detection through the synergy of statistical distance measurement and intelligent feature selection. Similarly, Biswakarma et al. [12] proposed an algorithm that combines the dual-stage Discrete Wavelet Transform and the median filter, which can effectively suppress noise interference, eliminate baseline drift, and extract 9 time-domain features from the QRS complex. Six different classifiers are used to perform classification on these features. The Gaussian kernel support vector machine (SVM) performs best on the MIT-BIH Arrhythmia Database, with a classification accuracy rate of 98.97%. However, its limitation mainly lies in the relatively high dependence on the accuracy of QRS segmentation. From a morphological perspective, Moghadam [13] presented an MI-detection framework that first extracts key morphological parameters from ECG waveforms and subsequently applies a random-forest classifier. While this approach yields promising diagnostic results on labeled data, it hinges on the robust localization of fiducial ECG points and has not been validated for noisy or multi-center environments. Collectively, these studies underscore both the potential and the remaining limitations of morphology-centric methods in real-world deployment.

2.2. ECG detection based on CNN

Ribeiro et al. [14] proposed an end-to-end DNN that automatically diagnoses six ECG abnormalities in 12-lead recordings, and its main strength lies in dispensing with hand-crafted features. The network, trained on two million equal-length strips, delivers high-precision single-label classifications. Nevertheless, a fixed-duration ECG usually contains several cardiac cycles and may exhibit multiple simultaneous abnormalities; assigning a single label therefore introduces noisy supervision that degrades both learning and evaluation. To address this limitation, Sadeghi et al. [15] introduced 3DECG-Net, which first fuses all 12 leads and then employs recurrence plots to transform the 1-D signals into 3-D representations. Within this richer space, ConvLSTM layers capture spatio-temporal patterns, while SE-residual blocks and multi-head attention jointly enable multi-label arrhythmia detection. The price of this expressiveness, however, is a heavy 3-D architecture that demands substantial computational resources and complicates deployment. Seeking a lighter yet more interpretable alternative, Chen et al. [16] proposed OSGAN, which leverages a biophysically inspired overlapping-grouping scheme and global-local feature fusion. By explicitly modeling hierarchical relationships among leads, OSGAN adaptively extracts full-scale features and simultaneously enhances both accuracy and explainability. Focusing specifically on atrial fibrillation (AF), Swetha et al. [17] devised the AB-LSTM model, whose CNN-BiLSTM-attention triple framework improves AF detection accuracy and robustness. In a complementary approach, Song et al. [18] presented HFFAM-BiLSTM, which combines heterogeneous feature fusion with an AM-DTW attention mechanism to accentuate discriminative ECG characteristics. After integrating variational mode decomposition (VMD) for noise suppression and tailored sample-balancing strategies, HFFAM-BiLSTM attains a clinical-level 97.1% accuracy on the AF detection task. [Table 1](#)

2.3. ECG detection with autoencoder

Kharzi et al. [19] introduced a real-time, embedded ECG anomaly detector that tackles the skin irritation and signal-quality degradation common to long-term monitoring with gel-based electrodes. The system first combined continuous wavelet transforms (CWT) with transfer learning; by converting the ECG signal into a 2-D scalogram, it greatly strengthened the expressive power of its features. Following this line of thought, Cao et al. [20] proposed a deep-learning-based model that classifies pathologies directly from these 2-D ECG images, detecting abnormal samples by automatically highlighting the most informative regions. Their pipeline employs Inception-v3 to capture global patterns, augments it with multi-channel bilinear attention and pooling to automatically highlight latent pathological regions, and refines the focus further through a multi-head attention module that dynamically re-weights each region. Although these fine-grained visual approaches achieve impressive accuracy, they incur high computational costs and offer only marginal gains in many practical anomaly-detection scenarios [21]. Recognizing this limitation, Shan [21] proposed ECGB_AAE, a hybrid model that fuses an adversarial autoencoder with a temporal convolutional network. By learning temporal representations exclusively from normal (positive) samples, the encoder-decoder pair yields

Table 1

Notation and definitions used in the proposed method.

Symbol	Description	Symbol	Description
x	Input ECG segment	T	Length of ECG segment
x_n	Normal ECG sample	τ	Anomaly detection threshold
Z	Latent variable	$\mathcal{L}_{\text{recon}}$	Reconstruction loss
μ	Mean of latent distribution	$\mathcal{L}_{\text{KL}}$	KL divergence loss
σ	Std. of latent distribution	LSTM	Long Short-Term Memory
Decoder	GRU	Encoder	VAE

reconstruction errors that are converted into a dynamic anomaly threshold, thereby eliminating reliance on scarce multi-class labels while still capturing subtle temporal deviations.

Roy et al.[22] introduced an unsupervised deep learning model developed ECG-NET an LSTM autoencoder. Employing a three-layer bidirectional LSTM encoder, the model compresses the ECG sequence into a low-dimensional latent space; a symmetric decoder then reconstructs the original signal. By training solely on normal ECGs, the method naturally mitigates class imbalance, while Kapur entropy-based automatic threshold selection on the reconstruction-loss distribution further boosts robustness against unseen anomalies. Building on this foundation, Roy et al.[23] proposed Attentive ECG RU, a new arrhythmia detector that swaps the LSTM backbone for a GRU autoencoder coupled with an attention mechanism. During inference, the reconstruction-loss boundary is no longer fixed; instead, a fuzzy threshold algorithm dynamically adapts it to regions where probability densities overlap, thereby overcoming the rigidity of a single cut-off. The

GRU architecture not only sidesteps the parameter redundancy inherent in LSTM cells but also lowers computational cost, and the fuzzy-logic threshold decision endows the model with greater tolerance to borderline cases.

Kalatehjari et al.[24] addressed the limitations of single-modal inputs in cardiovascular-disease diagnosis (CVD) by introducing a dual-modal CNN-BiLSTM model that jointly learns from ECG and phonocardiogram (PCG) signals. Synchronous analysis of the two modalities markedly improves the classification accuracy for coronary artery disease (CAD); nevertheless, the dual-branch CNN-BiLSTM architecture with the bilinear fusion layer incurs such high computational costs that deployment on resource-limited mobile medical devices remains impractical. Following a simpler single-channel approach, Gupta et al. [1] presented an automatic arrhythmia detector built solely on LSTM. Trained on 98,112 heartbeat segments, the model attains 98.85% accuracy, comfortably surpassing the competing CNN (92.11%), SVM (98.11%), and PCANet-SVM hybrid (97.94%).

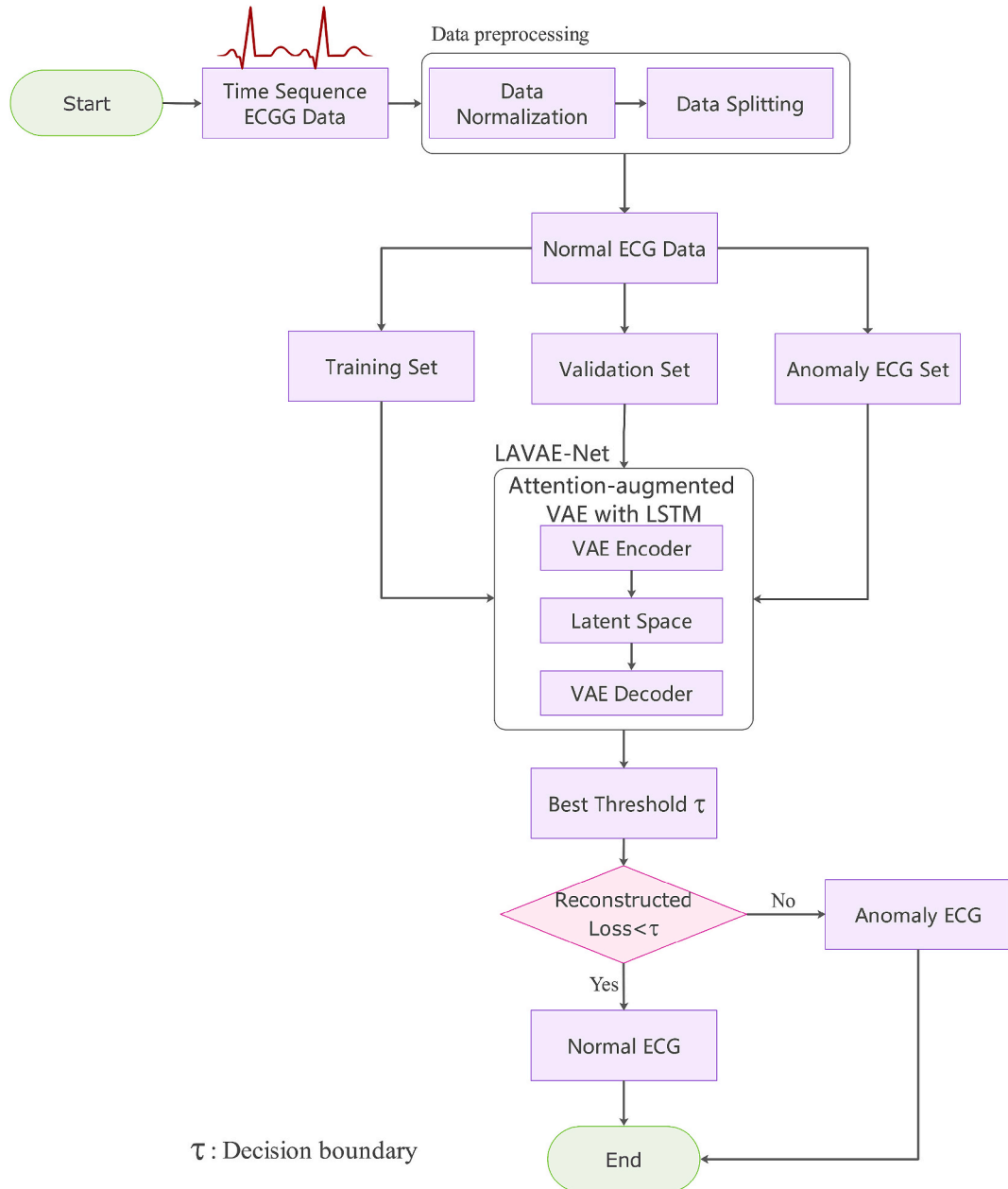


Fig. 1. Block diagram of ECG-VAE-NET architecture.

3. Notation and definitions

To facilitate readers' understanding of the proposed methodology, this section summarizes the key notations, variables, and abbreviations used throughout the method description.

4. Proposed method

4.1. Overall architecture

Fig. 1 presents the workflow of the proposed LAVAE-Net model, which employs an attention-augmented LSTM-based VAE for ECG anomaly detection. Initially, all samples are extracted from the ECG5000 dataset, normalized, and randomly shuffled before being divided into training, validation, and test sets according to a pre-determined ratio. During the training phase, only normal ECG data is used. The VAE encoder compresses the normal ECG signals into a low-dimensional latent space, while the decoder reconstructs the original ECG signals. After training, normal ECG signals exhibit minimal reconstruction errors, whereas anomalous ECG signals result in significantly higher reconstruction loss. Throughout the training process, a new threshold is generated every 10 epochs based on the training loss values

to distinguish between normal and anomalous ECG signals. The final threshold produced during training is selected as the optimal threshold for ECG prediction.

4.2. Model architecture

The proposed LAVAE model employs a VAE enhanced by an attention module that adaptively highlights critical temporal segments of the ECG signal, such as the P wave, QRS complex and T wave, to achieve unsupervised anomaly detection. Its core is a lightweight unidirectional LSTM-VAE whose capacity is further strengthened by this attention mechanism. The full architecture is presented in Fig. 2. In the encoder part, an ECG sequence of length t is first fed into a unidirectional LSTM, which outputs a hidden-state sequence of shape $[\text{batch_size}, \text{seq_len}, \text{embedding_dim}]$. Thereafter, Dropout and normalization are applied to regularize the activations, curb overfitting, and stabilize training. Building on this stabilized representation, the attention mechanism computes an importance weight for every time step; after softmax normalization, the resulting attention distribution is used to perform a weighted sum over the LSTM outputs, yielding a compact, fixed-length context vector. The linear layer maps the context vector to the mean parameter μ and logarithmic variance parameter $\log\sigma^2$ of the latent

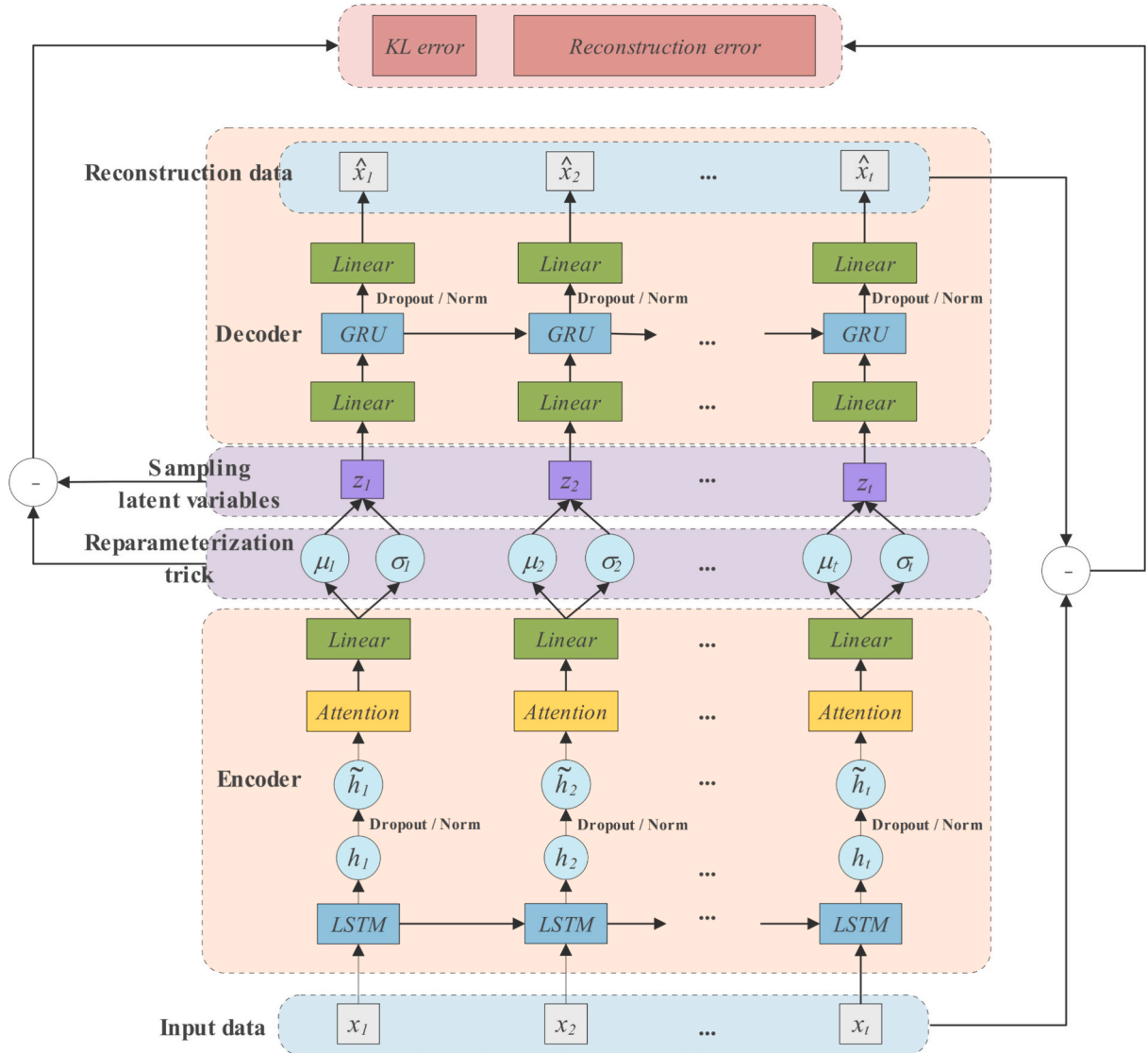


Fig. 2. Model architecture.

space respectively. These parameters define the posterior distribution $q(z|x)$ of the latent variables z . VAE Encoder (Encoder) is responsible for reconstructing the original ECG sequence from the latent space. Its structural parameters correspond to those of the Encoder. The Decoder receives the sampled latent variables sequence $\{z_1, z_2, \dots, z_t\}$ and maps it to the initialization information of the GRU through a linear layer. Unlike the encoder, which employs LSTM units, the decoder adopts the GRU structure; this choice reduces network complexity while preserving sequence-modeling capacity and simultaneously accelerates training. Leveraging the compressed information encoded in the latent variables, the GRU then progressively unfolds the reconstructed sequence step by step. After the output of the GRU at each time step passes through Dropout and normalization regularization, it is mapped to the reconstructed ECG signal value $\{\hat{x}_1, \hat{x}_2, \dots, \hat{x}_t\}$ through a linear layer.

The reconstruction error is calculated by comparing the original input and the reconstructed output. SmoothL1Loss is used to ensure robustness against the noise in the ECG signal. Meanwhile, the KL divergence constrains the latent distribution to be close to the standard normal distribution, preventing model degradation and maintaining the generative ability. These two loss components are combined through weighting to form the total loss, ensuring that the model can accurately reconstruct normal ECG patterns and maintain good generalization performance.

4.3. VAE encoder based attention

The role of the variational Encoder is to map the input ECG signal sequence through the mean vector and log variance vector to latent space representations and output the context vector of the attention layer. Overall, this process unfolds in three seamlessly connected stages: temporal feature extraction, normalization and attention aggregation, and latent distribution parameter mapping.

4.3.1. LSTM

To effectively capture the long-term dependencies in ECG signals, the input ECG sequence (with a length of 140 in the ECG5000 dataset) is fed into a unidirectional LSTM network. Leveraging its memory-cell structure, the network retains crucial historical information while simultaneously addressing the gradient-vanishing problem inherent in traditional RNNs. Moreover, opting for a unidirectional LSTM reduces the computational load by approximately 50% compared with a bidirectional counterpart, thereby rendering the system well-suited for deployment on embedded devices. Denoting the LSTM hidden-layer dimension as d , the corresponding update formulas are as follows:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (1)$$

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (2)$$

$$\tilde{c}_t = \tanh(W_g x_t + U_g h_{t-1} + b_g) \quad (3)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (4)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (5)$$

$$h_t = o_t \odot \tanh(c_t) \quad (6)$$

$i_t, f_t, \tilde{c}_t, c_t, o_t$ represent the input gate, forget gate, candidate cell state, current cell state, and output gate vector respectively.

c_t, h_t represent the hidden state at time t respectively; $W_i, U_i, W_f, U_f, W_o, U_o, W_g, U_g$ represent the training weight matrices of the corresponding gates and memory units respectively; b_i, b_f, b_o, b_g represent the learnable bias vector, and σ is the Sigmoid activation function.

After recursive iteration, the output sequence of the hidden state of the entire sequence at each time is:

$$H = \{h_1, h_2, \dots, h_T\}, \quad h_t \in R^d \quad (7)$$

The LSTM continuously accumulates long-term and short-term dependency information within the sequence, providing temporal features for subsequent attention mechanism and latent space projection.

4.3.2. Attention mechanism

To sharpen the model's focus on critical features, an attention mechanism is introduced; it heightens sensitivity to abnormal waveforms such as arrhythmia and ST-segment deviations while concurrently suppressing noise interference. Because clinical ECG recordings are easily contaminated, robustness is further strengthened by applying Dropout to prevent the network from over-relying on a handful of feature channels, and by using Layer Normalization to erase inter-subject amplitude differences and speed convergence. Specifically, a Dropout rate of 0.2 is adopted, and the hidden vector at every time step is layer-normalized, yielding the following set of normalized hidden-state vectors:

$$\tilde{H} = \{\tilde{h}_1, \tilde{h}_2, \dots, \tilde{h}_T\}, \quad \tilde{h}_t \in R^d \quad (8)$$

After the LSTM has produced its complete set of hidden states, a weighted context vector is derived via the unidirectional attention mechanism. This mechanism assigns an normalized attention weights to every element so that the timesteps most critical for identifying "Abnormal Waveforms" are emphasized. The algorithm proceeds through the following core steps:

1) Calculate an unnormalized score for the Hidden Vector at each time step as e_t :

$$e_t = w \tilde{h}_t + b, \quad e_t \in R \quad (9)$$

Among them, $w \in R^d, b \in R$ are the bias term parameters.

2) Apply softmax to all scores to get the attention weights:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{j=1}^T \exp(e_j)}, \quad \sum_{t=1}^T \alpha_t = 1 \quad (10)$$

3) Use a weighted sum to aggregate all normalized Hidden Vector into a context vector

$$c = \sum_{t=1}^T \alpha_t \tilde{h}_t, \quad c \in R^d \quad (11)$$

Through the mechanism of the Attention layer, the model can dynamically focus on local moments that may contain pathological features, avoiding information loss or long-range dependence decay caused by simply relying on the last hidden state.

4.3.3. Variational mapping and reparameterization

Variational mapping first compresses the high-dimensional data into the parameters of a low-dimensional probability distribution, thereby reducing dimensionality while preserving the data's essential features. To make this process differentiable and enable backpropagation, the reparameterization trick is then applied when sampling from the latent Gaussian distribution. The context vector $c \in R^d$ is mapped through two fully connected layers respectively to generate the mean vector $\mu \in R^{d_z}$ and the log-variance vector $\log \sigma^2 \in R^{d_z}$ of the latent space:

$$\begin{cases} \mu = W_\mu c + b_\mu, & W_\mu \in R^{d_z \times d}, b_\mu \in R^{d_z}, \\ \log \sigma^2 = W_{\log \sigma^2} c + b_{\log \sigma^2}, & W_{\log \sigma^2} \in R^{d_z \times d}, b_{\log \sigma^2} \in R^{d_z} \end{cases} \quad (12)$$

After the original ECG sequence of length T has successively passed through the LSTM, Dropout, LayerNorm, and Attention modules, it is compressed into a set of Gaussian-distribution parameter groups ($\mu,$

$\log\sigma^2$) of a specified dimension d_z . Leveraging the reparameterization trick, a latent vector z is then sampled from this distribution to serve the reconstruction task of the subsequent decoder. The formula is as follows:

$$z = \mu + \varepsilon \odot \exp\left(\frac{1}{2}\log\sigma^2\right), \quad \varepsilon \sim \mathcal{N}(0, I) \quad (13)$$

Among them, z is the sampled latent variable, μ is the mean of latent variables; $\log\sigma^2$ is the log variance of latent variables; ε denotes the noise drawn from the standard normal distribution with mean 0 and covariance as the identity matrix.

4.4. VAE decoder

The main function of the VAE Decoder is to map the low-dimensional latent vector z produced by the encoder back to the original time-series space, reconstructing a waveform that closely resembles the input ECG sequence. Because the model is trained exclusively on normal ECG data, the decoder learns to reproduce only normal morphologies, with their temporal dynamics and amplitude variations intact. In contrast, when an abnormal sample is presented, its latent representation typically lands on the periphery of the learned distribution, leading to a markedly higher reconstruction error. This selective difference in reconstruction ability forms the discriminant basis for anomaly detection: by setting a reconstruction error threshold, samples with a reconstruction error below the threshold are judged as normal, while those above the threshold are identified as abnormal.

4.4.1. Latent space mapping and GRU sequence generator

Map the latent vector $z \in R^{d_z}$ to the hidden layer dimension d of GRU through a fully connected layer to obtain the initial hidden state of Decoder:

$$h_z = W_z z + b_z, \quad W_z \in R^{d \times d_z}, \quad b_z \in R^d \quad (14)$$

Owing to the concise gating design and reduced parameter count of the GRU [25], it can efficiently capture both short-term and medium-term dependencies within ECG signals. The GRU update proceeds as follows:

$$r_t = \sigma(U_r h_{t-1} + b_r) \quad (15)$$

$$u_t = \sigma(U_u h_{t-1} + b_u) \quad (16)$$

$$\tilde{h}_t = \tanh(U_c(r_t \odot h_{t-1}) + b_c) \quad (17)$$

$$h_t = u_t \odot h_{t-1} + (1 - u_t) \odot \tilde{h}_t \quad (18)$$

$r_t, u_t, \tilde{h}_t, h_t$ are respectively Reset Gate, Update Gate, Candidate Hidden State, and Hidden State Update. U_r, U_u, U_c are the hidden state mapping weight matrices inside GRU and the bias vectors b_r, b_u, b_c inside GRU

4.4.2. Dropout and layerNorm

To alleviate overfitting and enhance training stability, the GRU hidden sequence $\{h_t\}$ is first obtained and then immediately subjected to Dropout and Layer Normalization, with the corresponding formulas given as follows:

$$\tilde{h}_t = m_t \odot h_t, \quad m_t \sim \text{Bernoulli}(1 - p) \quad (19)$$

$$\hat{h}_t^{(i)} = \gamma^{(i)} \frac{\tilde{h}_t^{(i)} - \mu_t}{\sqrt{\sigma_t^2 + \varepsilon}} + \beta^{(i)}, \quad \mu_t = \frac{1}{d} \sum_{i=1}^d \tilde{h}_t^{(i)}, \quad \sigma_t^2 = \frac{1}{d} \sum_{i=1}^d (\tilde{h}_t^{(i)} - \mu_t)^2 \quad (20)$$

γ, β among them are the scaling and offset parameters. In the system, it is 0.2.

4.4.3. Mapping from hidden state to feature space

The normalized GRU output is mapped back to the original feature

space to obtain the reconstruction of the input sequence, and the mapping formula is:

$$x_t = W_z h_t + \hat{b}_z, \quad W_z \in R^d, \quad \hat{b}_z \in R \quad (21)$$

x_t represents the output at time step t , which is the reconstructed input. h_t is the hidden state at time step t , output by the GRU. W_z is the weight matrix of the output layer, mapping the hidden state h_t to the original feature space. $W_z \in R^d$ means that the weight matrix, W_z has dimensions matching the feature space (where d is the dimensionality of the feature space). $\hat{b}_z \in R$ means the bias term is a scalar.

Finally, the reconstruction function is output as follows:

$$\hat{X} = \{\hat{x}_t\}_{t=1}^T \quad (22)$$

4.5. Loss function

The loss function $\mathcal{L}_{\text{VAE}}$ consists of two parts: $\mathcal{L}_{\text{recon}}$ (Reconstruction Loss) [26] and $\mathcal{L}_{\text{KL}}$ (Kullback-Leibler Divergence), and the formula is as follows:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KL}} \quad (23)$$

Among them, β is the balance factor (set to 0.1 in this paper) that harmonizes reconstruction fidelity with the strength of regularization, while the KL divergence, acting as the regularization term, ensures that the latent-variable distribution remains close to the standard normal distribution. In the standard VAE loss function, β is 1. We adopted the β coefficient multiplication $\mathcal{L}_{\text{KL}}$ proposed by Zhao [27] to achieve a balance between “reconstruction accuracy” and “latent distribution regularization”. Training the VAE on normal ECG data not only ensures high-fidelity reconstruction of normal sequences, but also encourages the latent space to approximate a standard normal distribution. As a result, when encountering abnormal samples, higher reconstruction errors will still occur due to the deviation from the expected distribution. To quantify the difference between the model's reconstructed output and the original input, the reconstruction loss is employed. In this paper, the Huber loss is used, and its formula is given as follows:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^N \sum_j \ell_\delta(x_j^{(i)} - \hat{x}_j^{(i)})^2 \quad (24)$$

N is the batch size, which represents the number of samples processed at one time; $x, \hat{x}$ are the input data and reconstructed data respectively. ℓ_δ is the Huber loss function;

$$\ell_\delta(a, b) = \begin{cases} \frac{1}{2}(a - b)^2 & \text{if } |a - b| < \delta \\ \delta|a - b| - \frac{1}{2}\delta^2 & \text{otherwise} \end{cases} \quad (25)$$

The formula for KL divergence is:

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^D \left(1 + \log\sigma_{ij}^2 - \mu_{ij}^2 - \sigma_{ij}^2\right) \quad (26)$$

Wherein, N is the batch size; D is the latent dimension; $\mu_{ij}^2, \sigma_{ij}^2$ are respectively the mean and variance of the j -th component of the latent variable mean for the i -th sample.

4.6. Automatic threshold estimation during training

4.6.1. Automatic threshold estimation

In ECG anomaly detection, the ECG waveforms are complex and vary greatly among individuals. Despite this complexity, normal heartbeats fall within a predictable range of variation. As the model progressively learns the characteristics of normal samples, its reconstruction

capability is enhanced, leading to a gradual reduction in the reconstruction error for these samples. The threshold, which distinguishes normal from abnormal samples, should be updated dynamically to track the model's improving reconstruction ability and maintained at the optimal quantile. In this paper, rather than relying on a static threshold [22], we update the threshold every 10 epochs during training.

For each validated time series x , the model gives the reconstructed output $\hat{x}$. The sum of squared errors is used, and the formula is:

$$e(x) = \sum_{t=1}^T (x_t - \hat{x}_t)^2 \quad (27)$$

Collect the set of error values from all normal verification samples $\{e_1, e_2, \dots, e_N\}$. Given that the expected proportion of abnormal samples is α , the quantile of $1-\alpha$ minus is taken as threshold τ :

$$\tau = \text{Quantile}\{e_i\}_{i=1}^N (1 - \alpha) \quad (28)$$

Following each set of 10 training rounds, abnormality is assessed using the current threshold τ , with the assessment formula being:

$$e(x_{\text{new}}) > \tau \quad (29)$$

During the early training stage, when the model experiences significant fluctuations, a fixed threshold might lead to an excessively high misjudgment rate. A dynamic threshold helps to buffer the error fluctuations during the model's immature stage and prevents the model from over-optimizing specific threshold points in later stages.

4.6.2. Effect of N -Epoch threshold updates

To evaluate the impact of threshold update frequency on the model's performance, we conducted experiments where the threshold was updated at intervals of every N epoch (with N values of 5, 10, 20, and 30). The performance metrics, including Precision, Recall, Specificity, F1-score, FPR, and FNR, were calculated for each setting, as summarized in Table 2.

Table 2 shows the threshold behavior throughout the training process for different update intervals. The results indicate that updating the threshold every 10 epochs achieves the best balance between performance and stability, as reflected by the highest Precision, Recall, and F1-score, while minimizing FPR and FNR.

Fig. 3 shows the threshold update behavior, tracking its change over the entire training course at different update frequencies of $N = 5, 10$, and 20 epochs after Savitzky-Golay filtering. The 10-epoch interval achieves the optimal balance between adaptability and stability. It allows sufficient flexibility to adjust the threshold in response to evolving data, while minimizing excessive fluctuations. This interval ensures that the model remains adequately sensitive to abnormal ECG patterns, while preventing overfitting to noise or anomalies that do not reflect true abnormalities. Therefore, we set the threshold update frequency to 10 epochs for our experiments.

4.6.3. Selection of the expected abnormal sample ratio

In our approach, the expected proportion of abnormal samples α is set to 0.05 based on prior experience in anomaly detection and typical distributions of abnormal events in ECG data. In many real-world scenarios, the occurrence of anomalies is rare, and abnormal ECG patterns are expected to be infrequent compared to normal beats. Therefore, we

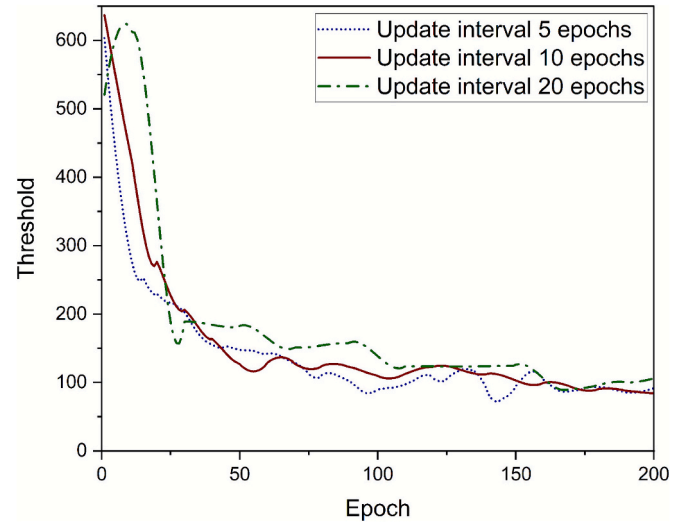


Fig. 3. Threshold update behavior for different update intervals.

empirically selected $\alpha = 0.05$ as a reasonable assumption for the abnormal sample ratio, aligning with prior studies and clinical knowledge about the rarity of such events in the general population.

5. Experiment

This section details the public dataset ECG5000, dataset analysis, dataset partitioning, and evaluation metrics.

5.1. ECG dataset

This study employs the publicly available ECG5000 dataset (<https://www.timeseriesclassification.com/description.php?Dataset=ECG5000>). The ECG5000 dataset is a benchmark dataset derived from the MIT-BIH Arrhythmia Database hosted on PhysioNet. It comprises 5,000 single heartbeat excerpts extracted from 20-hour ambulatory ECG recordings, preprocessed and standardized by the UCR Time Series Classification Archive [25]. Each sample consists of 140 time points (sampled at 125 Hz, spanning ≈ 1.12 s) and has undergone noise reduction, baseline correction, and amplitude normalization. ECG5000 serves as a key benchmark for evaluating heartbeat classification algorithms (e.g., CNNs, RNNs), anomaly detection methods, and lightweight embedded medical diagnostic systems. The types and quantities of ECG contained in the dataset are as shown below.

5.1.1. Dataset analysis

Table 3 shows a significant imbalance among ECG types, where normal sinus rhythm accounts for 58.38% of the dataset. The mean and standard deviation curves for each type of ECG signal are depicted in Fig. 4. The solid curves represent mean values, and the shaded areas show waveform fluctuation ranges. It is clear that there are notable differences in temporal characteristics between normal and abnormal signals. Normal waveforms exhibit a certain regularity, with segments including the P wave, QRS complex, and T wave, which have relatively fixed morphologies. However, distortions may occur due to

Table 2
Effect of threshold update frequency on model performance.

Update Interval K (epochs)	Precision	Recall (Sensitivity)	Specificity	F1-score	FPR	FNR
5	0.9848	0.9942	0.9452	0.9895	0.0548	0.0058
10	0.9829	0.9966	0.9384	0.9897	0.0616	0.0034
20	0.9843	0.9957	0.9435	0.9900	0.0565	0.0043
30	0.9848	0.9952	0.9452	0.9900	0.0548	0.0048

Table 3
Category types of ECG5000.

Category types	Quantity
Normal beats	2919
R-on-T premature ventricular contractions (R-on-T PVC)	1767
Premature ventricular contractions (PVC)	96
Supraventricular premature beats (SP)	194
Unclassified beats (UB)	24
Total	5000

abnormalities such as ventricular tachycardia or premature ventricular contractions. Conversely, abnormal heartbeats deviate from the normal pattern, exhibiting different regular changes. Through time-domain analysis of the ECG5000 dataset, significant morphological differences between normal and abnormal data are observed. These morphological differences suggest potential for automated anomaly detection.

5.1.2. Dataset split

As illustrated in Fig. 5, the dataset was divided into training and evaluation components. Specifically, 80% of the normal ECG data was allocated for training purposes, while the remaining 20% of the normal data, combined with all the abnormal data, was set aside for evaluation and testing. Regarding the validation set, we clarify that it is composed exclusively of normal ECG samples, similar to the training set. This ensures that the model's ability to generalize to unseen normal data is evaluated effectively during the training process.

5.2. Experimental setup parameters

The experiments were carried out on a computer system featuring an Intel Core i7-10700KF CPU, 32 GB of RAM, and an NVIDIA GeForce RTX 4070Ti Super GPU with 8 GB of VRAM. The implementation was conducted within a Python 3.10 environment, leveraging CUDA 11.8 and the PyTorch 2.5.1 framework to develop the methods described in this paper. The training hyperparameters used in this study are summarized in Table 4.

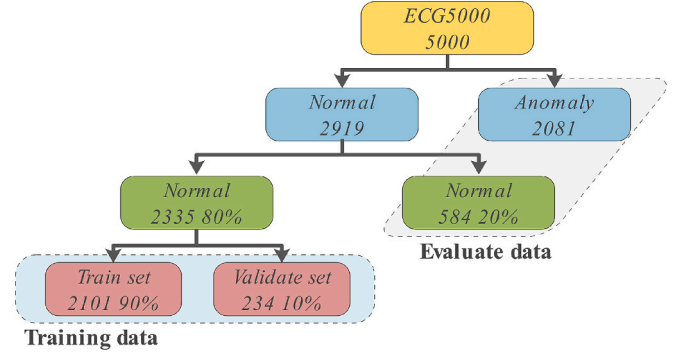


Fig. 5. Dataset split.

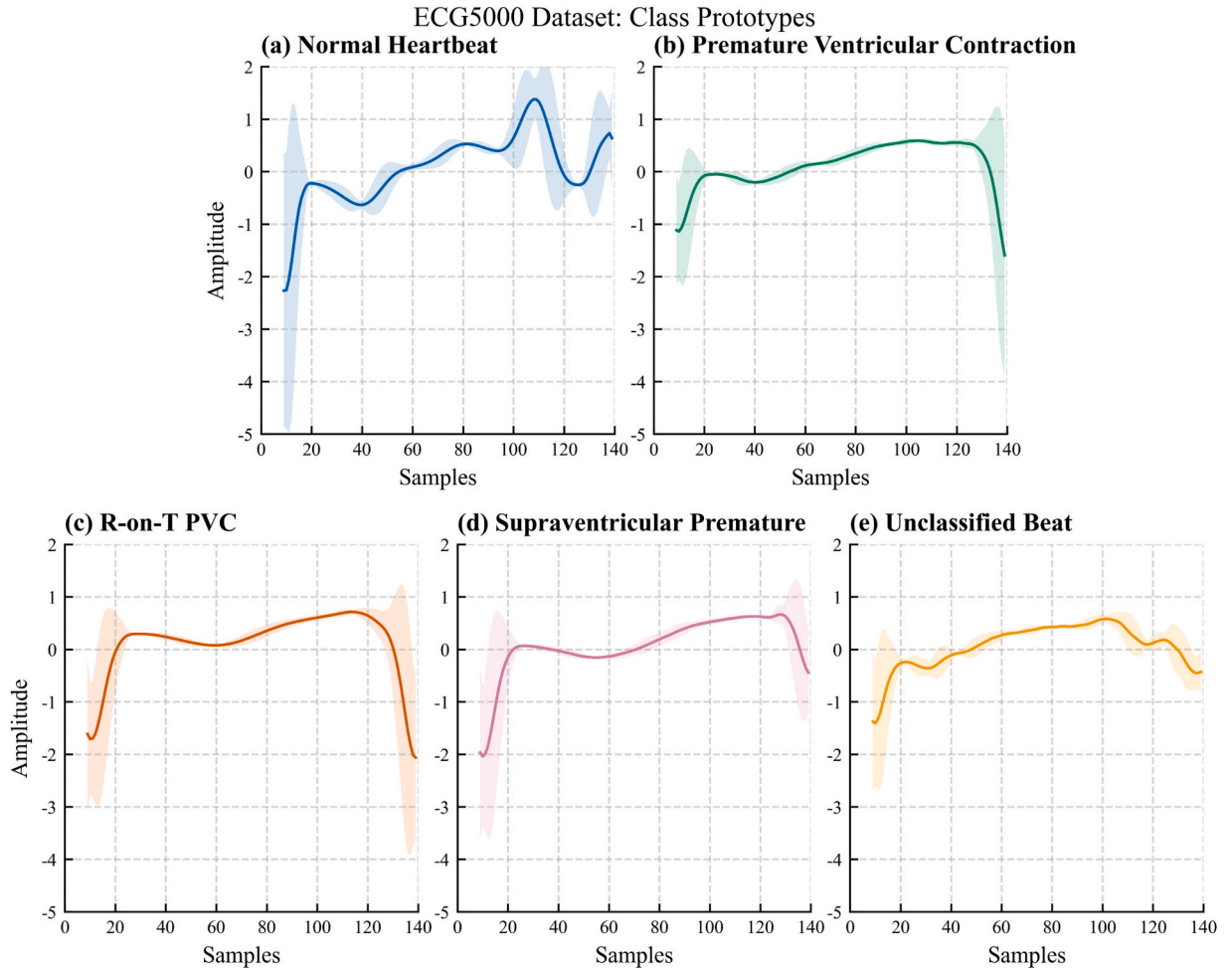


Fig. 4. Morphological variability graph of each ECG class.

Table 4
Training hyperparameters.

Parameters	Values
Optimizer	Adam
Learning rate	0.001
Batch size	8
Hidden layer size of LSTM	64
Latent variable dimension	32
Iterations	500
Dropout rate	0.2
Expected Abnormal Sample Ratio Selection	0.05
KL Weight	0.1

5.3. Hyperparameter sensitivity analysis

The embedding dimension (EMBEDDING_DIM) parameter controls the dimensionality of the hidden states in the LSTM layer and attention mechanism. A larger embedding dimension may allow the model to capture more features and complex patterns; however, it could also lead to higher computational complexity and an increased risk of overfitting. On the other hand, a smaller embedding dimension helps reduce model complexity and training time, but it may not fully capture all the details in the input data. The latent dimension (LATENT_DIM) parameter controls the size of the latent space in the VAE. A larger latent dimension enables the model to encode more complex data structures but may lead to redundancy in the latent space, which could affect the model's generalization ability. A smaller latent dimension limits the model's representational capacity but may help improve generalization and training efficiency. The impact of different parameters on model performance, with all other parameters kept constant, is as Table 5.

From Table 5, it can be observed that the second configuration (Embedding Dim = 64, Latent Dim = 32) performs better due to several key factors. Firstly, it achieves a higher recall (sensitivity) of 0.9966, compared to 0.9885 in the first configuration. This suggests that the model with the second configuration is more effective at identifying actual positive cases, thereby reducing the number of missed positives. Although the precision of the second configuration (0.9862) is slightly lower than the first (0.9875), the difference is minimal, indicating that the second configuration still performs well in terms of accurately identifying positive instances.

Moreover, the F1-score of the second configuration is slightly higher (0.9914) compared to the first (0.9880), reflecting a better balance between precision and recall. This improvement in the F1-score demonstrates the second configuration's enhanced overall performance. While the specificity and false positive rate (FPR) are marginally lower in the second configuration, these differences are minimal and do not significantly affect its overall effectiveness. Additionally, the second configuration has a notably lower false negative rate (FNR) of 0.0034, compared to 0.0115 in the first configuration. This reduction in false negatives indicates that the second configuration is more capable of correctly identifying positive cases, which is especially important in applications such as anomaly detection.

5.4. Performance evaluation metrics

This study utilizes Accuracy, Precision, Recall, and F1-score as the evaluation metrics. Accuracy refers to the percentage of ECG signals that are correctly classified out of the entire test dataset. Precision is defined

as the ratio of the number of samples correctly identified as class i to the total number of samples predicted as class i . Recall, on the other hand, measures the proportion of samples that the model correctly identifies as class i relative to all samples that actually belong to class i . The F1-score is a comprehensive metric that takes both precision and recall into account, providing a balanced measure especially useful for imbalanced datasets. The formulas for these metrics are as follows:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (30)$$

$$Precision = \frac{TP}{TP + FP} \quad (31)$$

$$Recall = \frac{TP}{TP + FN} \quad (32)$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (33)$$

Where TP is the number of normal points correctly detected, FP is the number of anomaly points incorrectly identified as normal points, and FN is the number of normal points incorrectly identified as anomaly points.

5.5. Experiment results

Collectively, the training and validation loss curves maintain parallel trajectories throughout the training process, with their divergence consistently bounded within a 5–15% range. The absence of significant divergence or rebound phenomena indicates no substantial overfitting in the model. This section presents the training and validation outcomes of the LAVAE-Net model. Fig. 6 illustrates the progression of the training and validation loss throughout the training process. The initial 20 epochs encompassed a typical “warm-up” phase coupled with the primary feature learning stage. During this period, the training loss rapidly declined from approximately 140 to around 40, and the validation loss decreased from about 128 to roughly 60, marking a reduction of over 50% for both. The steep descent in these curves signifies the model's swift acquisition of key features from normal ECG samples via gradient updates. As training progressed beyond the 100th epoch, the training loss curve began to flatten, gradually descending to about 32, while the validation loss also progressively converged towards approximately 38, achieving its minimum by the 160th epoch. The concordant decrease in both loss curves indicates ongoing performance enhancement through fine-tuning after substantial feature learning. However, the comparatively smaller decrease in validation loss suggests the model was approaching its performance plateau.

The validation loss curve exhibits short-term, high-frequency fluctuations (e.g., around the 50th, 100th, and 150th epochs), with peaks exceeding the adjacent level by approximately 10–15%. This phenomenon could primarily stem from statistical variability in mini-batch sampling during evaluation. To mitigate such jitter, consider increasing the validation batch size, fixing the random seed, or applying a moving average to the assessment results to obtain a smoother and more reproducible loss curve.

Collectively, the training and validation loss curves maintain parallel trajectories throughout the training process, with their divergence consistently bounded within a 5–15% range. The absence of significant

Table 5
Impact of embedding dimension and latent dimension on model performance.

Embedding dim	Latent dim	Precision	Recall (Sensitivity)	Specificity	F1-score	FPR	FNR
32	16	0.9875	0.9885	0.9555	0.9880	0.0445	0.0115
64	32	0.9862	0.9966	0.9503	0.9914	0.0497	0.0034

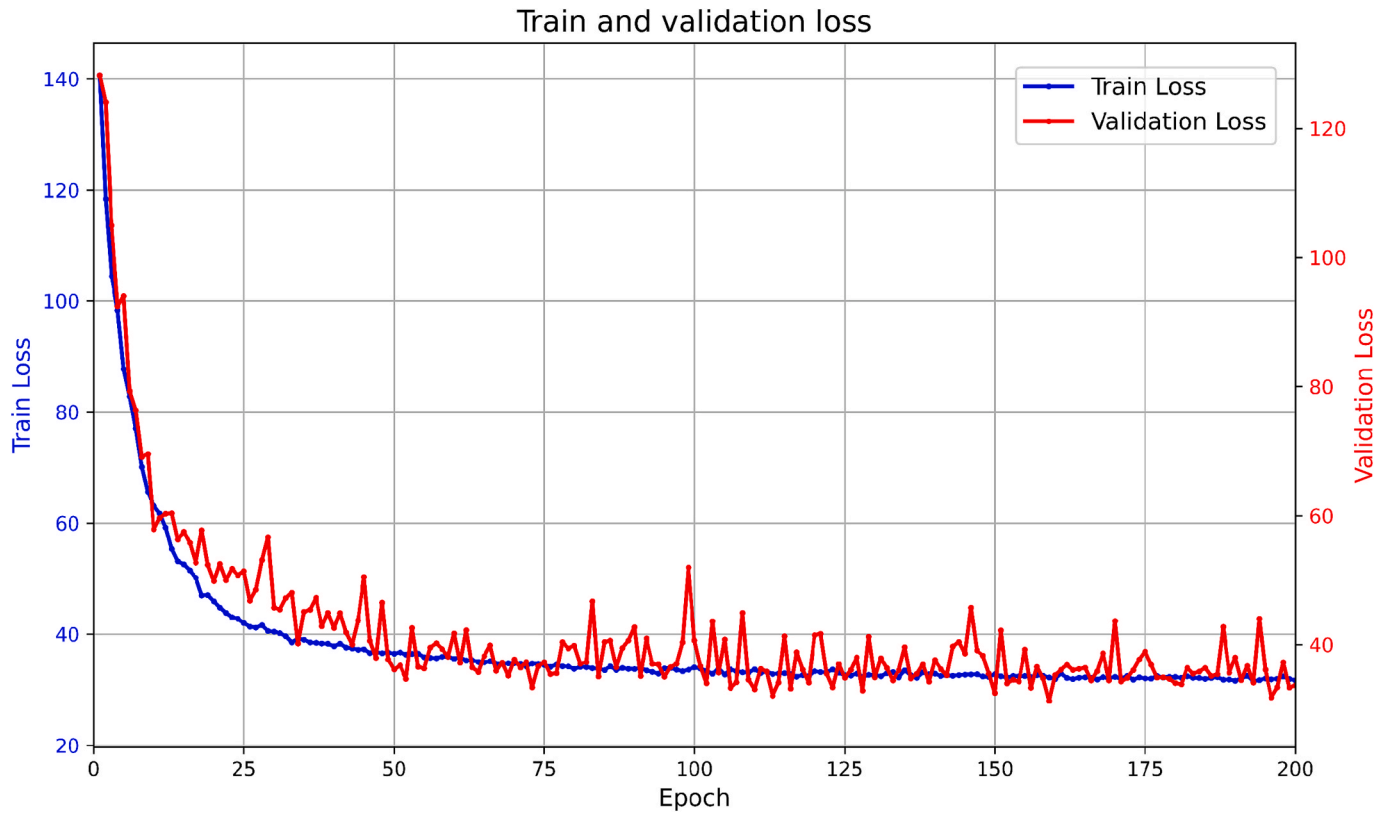


Fig. 6. Training and validation losses.

divergence or rebound phenomena indicates no substantial overfitting in the model.

Tables 6 and 7 jointly present the confusion matrix and the quantitative performance metrics for the test set. After training the model solely on normal ECG signals, we achieved efficient anomaly detection on the ECG5000 dataset. The test data comprised 584 normal records and 2,081 arrhythmia records. As a result, the model attained an overall accuracy of 99.06%, which implies that only about 0.94% of samples were misclassified. More precisely, the false-negative rate was as low as 0.43%, whereas the false-positive rate stood at 2.74%. The false-negative rate underscores the model's robustness, thereby meeting the stringent dual-control requirements for both false alarms and missed detections in clinical monitoring. Nevertheless, the model still exhibits limited sensitivity to certain subtle abnormal patterns. Furthermore, the F1-Anomaly score of 99.04% highlights the balanced trade-off between precision and recall for anomaly samples. Specifically, the model accurately identifies the vast majority of anomalies while simultaneously keeping the false-positive rate at an extremely low level. The model demonstrates the ability to efficiently detect ECG abnormalities while effectively suppressing false alarms from normal samples, indicating its potential for deployment in clinical settings and remote monitoring scenarios where stringent requirements exist for both false alarms and missed detections.

Table 6
Confusion matrix for testing set.

		Predicted class	
		Normal	Anomaly
True class	Normal	568	16
	Anomaly	9	2072

Table 7
LVAE-Net performance.

Evaluation metrics	Value
Accuracy	99.06%
Precision _(Normal)	97.26%
Recall _(Normal)	98.44%
F1-Score _(Normal)	97.85%
Precision _(Anomaly)	99.57%
Recall _(Anomaly)	99.56%
F1-Score _(Anomaly)	99.40%

5.6. Noise robustness evaluation

To evaluate the robustness of the proposed LVAE-Net under realistic signal corruption, we conducted a systematic noise robustness experiment by injecting synthetic yet physiologically plausible artifacts into ECG signals during inference. Importantly, the model was trained only on clean normal ECG5000 signals following the same protocol as described in the previous experiments, and no retraining or threshold recalibration was performed for any noise condition. This setting reflects a realistic deployment scenario in which a pre-trained model encounters noisy signals in practice.

5.6.1. Noise modeling and injection

To simulate realistic signal corruption encountered in practical ECG monitoring scenarios, we performed a noise robustness study by artificially injecting representative physiological artifacts into the ECG signals at the inference stage. Importantly, the proposed LVAE-Net was trained exclusively on clean normal ECG samples, and no retraining, fine-tuning, or noise-specific threshold adjustment was applied during the robustness evaluation. This design reflects a realistic deployment setting in which a pre-trained anomaly detection model must operate reliably under varying noise conditions.

Three common categories of ECG noise were considered in this study. First, baseline wander was introduced to emulate low-frequency signal drift caused by respiration and gradual electrode impedance changes. This effect was simulated by adding a sinusoidal waveform with a frequency of 0.3 Hz, whose amplitude was scaled proportionally to the standard deviation of the original ECG signal, thereby preserving subject-dependent signal characteristics. Second, EMG-related noise, representing high-frequency interference from skeletal muscle activity, was modeled as additive Gaussian white noise. Different noise intensities were controlled using signal-to-noise ratios (SNRs) of 20 dB and 15 dB, corresponding to moderate and relatively severe muscle interference, respectively. Third, motion artifacts, which commonly arise from abrupt electrode displacement or patient movement, were simulated by injecting short-duration, high-amplitude Gaussian pulses at random temporal locations along the ECG sequence.

In addition to individual noise sources, a combined noise condition was constructed by superimposing baseline wander, EMG noise, and motion artifacts within the same ECG segment. This setting was designed to approximate challenging real-world environments in ambulatory or wearable ECG monitoring, where multiple noise sources often coexist. All noise injection procedures were applied independently to each ECG segment at the test stage, while the underlying temporal structure and length of the signals were preserved. By maintaining consistent preprocessing and evaluation protocols across all noise conditions, this experiment enables a fair and controlled assessment of the intrinsic robustness of the proposed model.

5.6.2. Results and analysis

Table 8 summarizes the robustness performance of LAVAE-Net under different noise conditions. Compared with the clean-signal results reported in Section 4.5, the proposed model exhibits a moderate and gradual performance degradation as noise severity increases, indicating stable behavior under realistic signal corruption.

For baseline wander and EMG noise, the model maintains strong anomaly detection capability, with F1-scores remaining close to those obtained under clean conditions. This suggests that the learned temporal representations are largely insensitive to low-frequency drift and moderate high-frequency interference. Even under increased EMG noise intensity, the reduction in performance is limited, demonstrating effective suppression of muscle-related perturbations.

Motion artifacts introduce a more noticeable impact on normal-sample specificity, leading to a decrease in normal accuracy. Nevertheless, anomaly accuracy remains consistently high, indicating that the model continues to reliably identify abnormal ECG patterns. This behavior reflects a conservative detection tendency under abrupt disturbances, prioritizing sensitivity over specificity, which is desirable in safety-critical monitoring scenarios.

Under the combined noise condition, where multiple artifacts coexist, the model preserves a balanced performance profile without catastrophic failure. Overall, these results confirm that LAVAE-Net demonstrates graceful degradation rather than abrupt performance collapse, supporting its practical applicability in noisy real-world ECG monitoring environments.

Table 8
Performance of LAVAE-under different noise conditions.

Noise condition	Normal Accuracy	Anomaly Accuracy	F1-score	AUC
Baseline Wander	0.9297	0.9980	0.9892	0.9852
EMG Noise (20 dB)	0.9452	0.9961	0.9904	0.9919
EMG Noise (15 dB)	0.9280	0.9975	0.9888	0.9899
Motion Artifact	0.8938	0.9985	0.9546	0.9797
Combined Noise	0.9315	0.9966	0.9887	0.9887

5.7. Latent space visualization and analysis

Two-dimensional projections of the latent mean vectors learned by the VAE encoder using t-SNE and UMAP are shown in Fig. 7. In Fig. 7(a), normal ECG samples form a compact and coherent cluster in the latent space, whereas abnormal samples are distributed over a substantially broader region and tend to lie outside the high-density normal manifold. Although a small number of abnormal samples appear near the boundary of the normal cluster, the overall separation indicates that abnormal patterns deviate systematically from the learned representation of normal cardiac dynamics.

Consistent observations are obtained from the UMAP projection in Fig. 7(b). Normal samples occupy a dense, low-variance region, while abnormal samples extend along multiple directions and populate peripheral or low-density areas of the latent space. The agreement between t-SNE and UMAP suggests that the observed structure is not an artifact of a specific dimensionality reduction technique, but rather reflects an intrinsic organization of the learned latent representation.

These results provide intuitive evidence that the proposed VAE learns a structured and continuous latent manifold for normal ECG signals, with anomalies naturally residing outside the core region of the normal distribution. This latent-space geometry is well aligned with the reconstruction-error-based anomaly detection strategy adopted in this work.

5.8. Interpretability and attention visualization

For each ECG class, a representative beat was selected and fed into LAVAE-Net using a forward pass that outputs the encoder attention weights, as shown in Fig. 8.

Fig. 8 presents the attention-weighted reconstruction analysis for four representative anomalous ECG beats from the ECG5000 dataset, including R-on-T, PVC, SP, and UB. Each subplot overlays the raw ECG waveform (black) with the normalized attention weights (blue dashed line), while the background heatmap reflects the temporal distribution of encoder attention. Higher attention regions correspond to segments that contribute more strongly to the learned latent representation and, consequently, to anomaly detection. (I) For the R-on-T case, the model assigns dominant attention to the premature R wave overlapping the preceding T-wave repolarization, capturing the electrophysiological hallmark of the R-on-T phenomenon that is known to trigger malignant arrhythmias. (II) In standard PVC beats, attention is primarily concentrated on the broad and distorted QRS complex, indicating that the model accurately focuses on the aberrant ventricular depolarization characteristic of ectopic ventricular activity. (III) In contrast, SP (supraventricular premature) beats show an early-shifted attention distribution focused on the P-wave region and the onset of the cardiac cycle, corresponding to premature atrial activation rather than ventricular origin. (IV) The UB class exhibits a more dispersed attention pattern, covering both early irregular oscillations and pronounced terminal distortions, consistent with its heterogeneous or atypical morphology. Collectively, these visualizations demonstrate that the temporal segments emphasized by the attention mechanism align closely with clinically meaningful electrophysiological features.

6. Discussion

6.1. Data normalization

To eliminate dimensional differences in sensor data, Z-score normalization is applied during preprocessing. The data are transformed to a standard normal distribution (mean = 0, std = 1), which enhances model convergence and stability through uniform feature scaling. The normalization formula is:

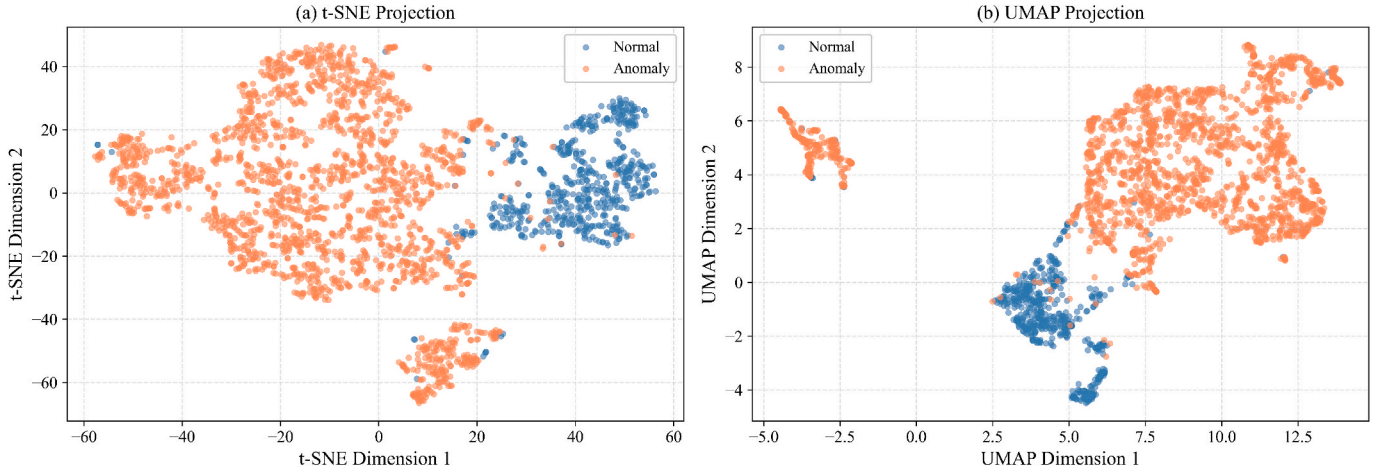


Fig. 7. Latent space visualization of normal and abnormal ECG samples.

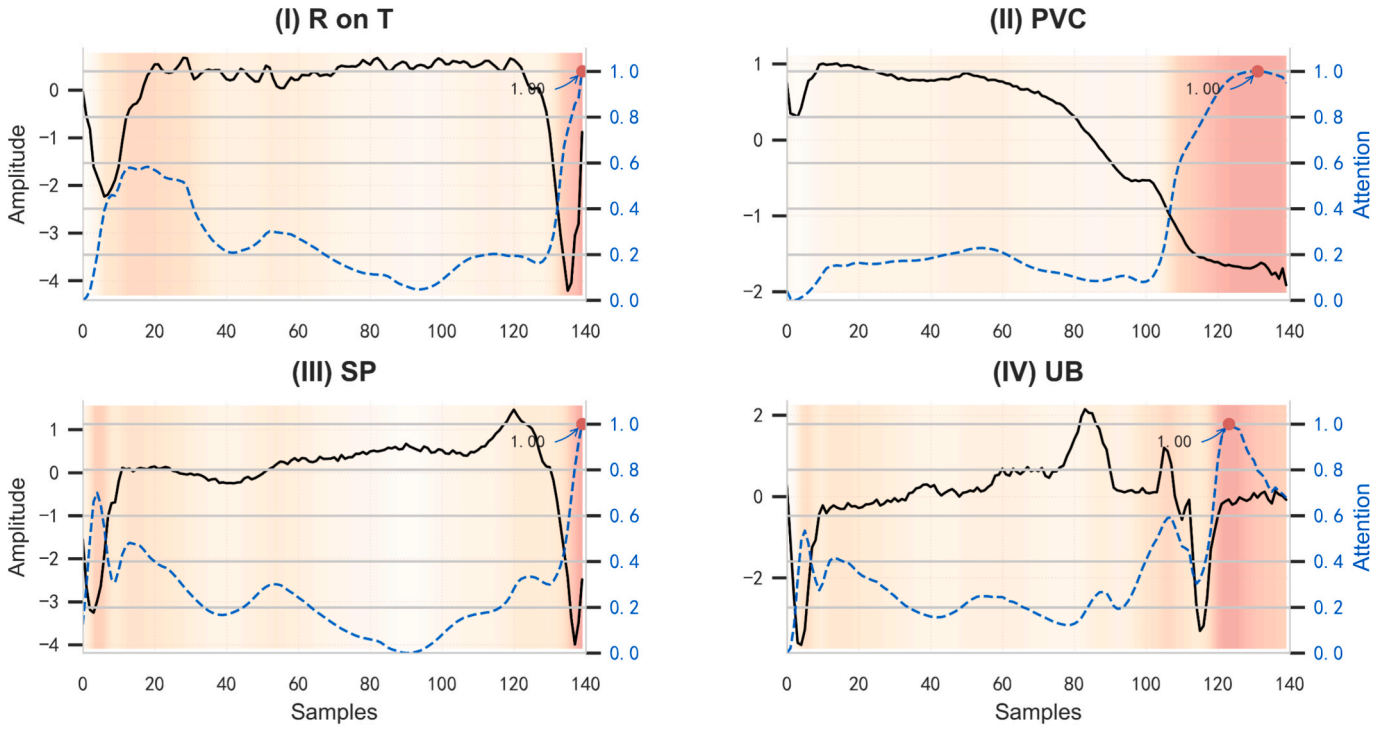


Fig. 8. Comparison across four abnormal ECG types.

$$\mu_j = \frac{1}{N} \sum_{i \in \mathcal{I}'} x_{ij} \quad (34)$$

$$\sigma_j = \sqrt{\frac{1}{N} \sum_{i \in \mathcal{I}'} (x_{ij} - \mu_j)^2} \quad (35)$$

$$x'_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (36)$$

Where, μ_j is the mean of sample j , σ_j is the standard deviation of sample j . and x'_{ij} is the Z-score normalized value of sample j .

Roy [22] noted that the autoencoder in their study did not use normalization. To evaluate its impact, they trained the model with identical hyperparameters on both raw and Z-score-normalized data, testing both versions on the same test set. The accuracy results are shown in Fig. 9.

The comparative analysis demonstrates that standard scaler pre-processing significantly enhances detection performance: accuracy improves from 80.83% to 98.57%, and recall rises substantially from 53.35% to 98.23%.

6.2. Model advantages

This section examines the advantages of the LAVAE-Net model by analyzing loss distributions across three distinct model architectures.

6.2.1. Comparison of loss distributions between normal and abnormal samples

To benchmark the proposed approach, we implemented three architectures and trained them with identical hyperparameters on the same dataset while evaluating their Kernel Density Estimation.

Method 1, Autoencoder [22], employs a three-layer LSTM autoencoder for unsupervised anomaly detection and updates its decision

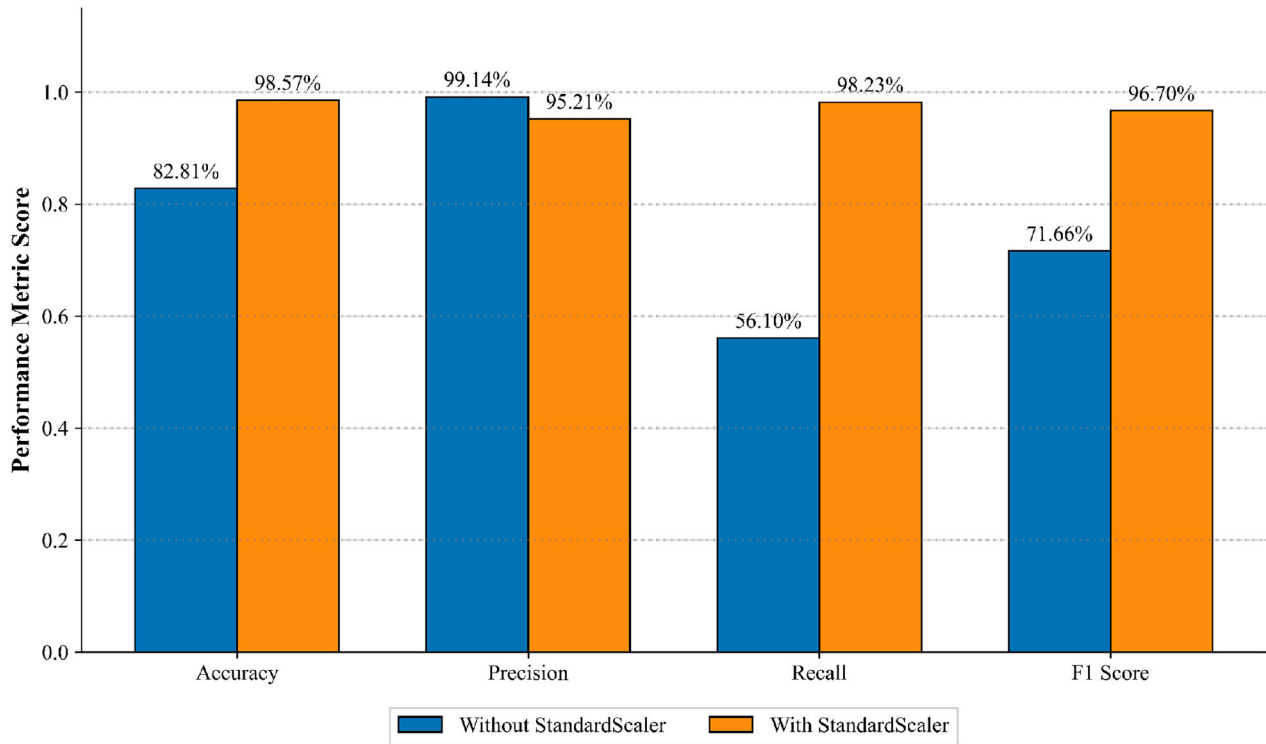


Fig. 9. Impact of feature standardization on classification performance.

boundary using the Kapur threshold strategy after every epoch. Kapur's thresholding method is employed as an automated approach for selecting the reconstruction loss threshold. Originally used for image segmentation, this method is adapted to work with the reconstruction losses of both normal and abnormal ECG signals. The process begins by computing the reconstruction loss for each test sample, both normal and abnormal. These losses are then used to construct a histogram that represents their frequency distribution.

Method 2, LSTM-VAE [26], combines LSTM with a VAE to perform multimodal anomaly detection; it was originally designed to identify abnormal motions during robotic assisted feeding in order to reduce

operational risk.

Method 3, LAVAE-Net, introduces probabilistic latent space representations within the VAE, strengthening the model's ability to model data distributions and improving the separation between normal and abnormal patterns.

Figs. 11 and 12, which employ VAE encoding, set the abnormality lower bound at 80–100, markedly higher than the threshold of 50 used by the Autoencoder in Fig. 10. In Fig. 12, LSTM-VAE-Attention produces a tighter abnormal distribution whose error peaks cluster between 200 and 250, while the normal-distribution peak remains well below the threshold; consequently, almost all abnormal errors lie above the cut-off

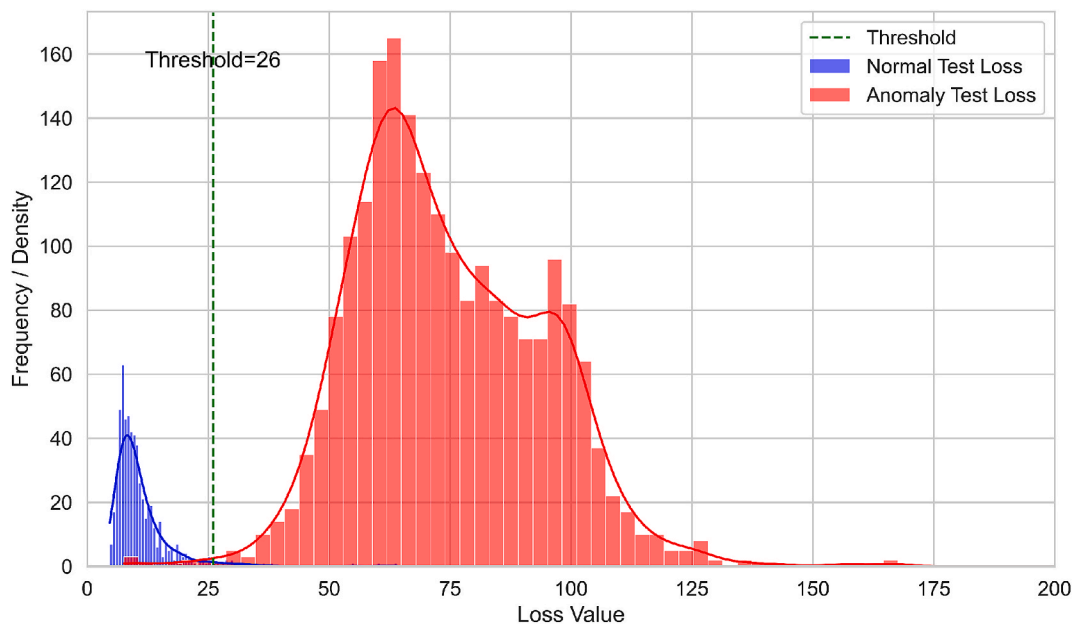


Fig. 10. Autoencoder: Test loss distribution with KDE.

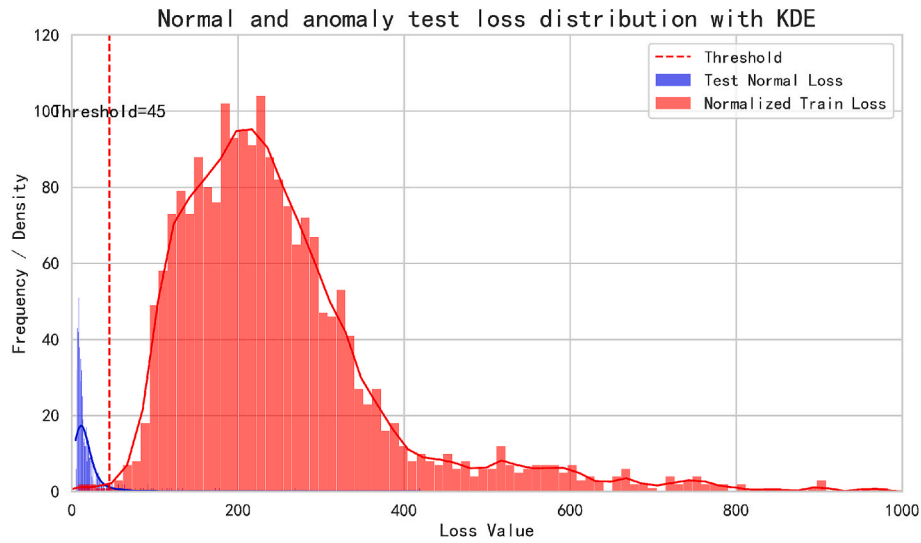


Fig. 11. LSTM-VAE: Test loss distribution with KDE.

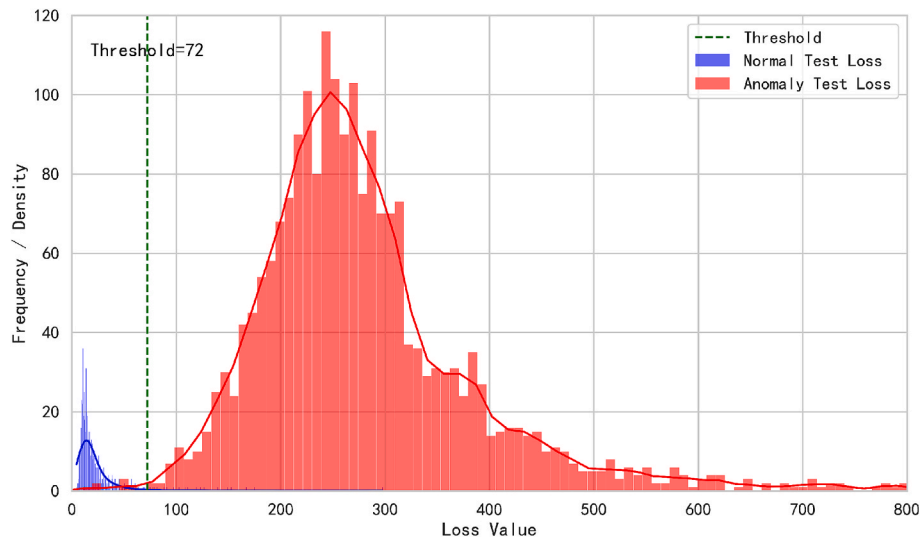


Fig. 12. LAVAE-Net: Test loss distribution with KDE.

and the long tail is suppressed. This pattern demonstrates that the attention layer is more sensitive to local distortions such as QRS and T-wave deviations and can capture subtle shifts caused by mild arrhythmias.

6.2.2. Ablation study

To quantify the individual contributions of the LSTM encoder, attention mechanism, and VAE latent modeling, we conducted an ablation study based on four model variants trained under identical settings, as shown in Table 9.

In Table 9, LSTM-VAE-Attention achieves 99.06% overall accuracy,

97.26% precision, 98.44% recall, and 97.85% F1-score on the ECG5000 test set. The attention mechanism improves feature localization by emphasizing diagnostically relevant ECG segments (e.g., QRS region, repolarization deviations). This reduces false positives, which primarily increases precision, while also providing marginal gains in recall. Specifically, accuracy rises by 1.35% and 0.71% relative to Autoencoder and LSTM-VAE, respectively; precision improves by 3.94% over LSTM-VAE; and the F1-score gains 3.23% and 1.73%, indicating simultaneous reductions in false negatives and false positives. Moreover, the sharper, taller abnormal distribution of LAVAE-Net simplifies threshold selection.

By integrating LSTM to model temporal dynamics and attention to emphasize critical waveform segments, LAVAE-Net automatically focuses on diagnostically relevant regions such as QRS complexes and T-waves. More importantly, the VAE component imposes probabilistic regularization on the latent space, ensuring a smooth and structured representation that separates normal physiological variability from abnormal deviations. This three-way synergy of LSTM-based temporal encoding, attention-driven feature localization, and VAE-enabled latent distribution modeling allows the network to precisely reconstruct global ECG morphology while amplifying subtle local abnormalities. As a

Table 9

Performance comparison of different models.

Metrics	Autoencoder (Only)	Autoencoder-Attention	LSTM-VAE (without attention)	LAVAE-Net (LSTM-VAE-Attention)
Accuracy	0.9771	0.9804	0.9835	0.9906
Precision	0.9178	0.9341	0.9332	0.9726
Recall	0.9763	0.9815	0.9909	0.9844
F1-Score	0.9462	0.9572	0.9612	0.9785

result, LAVAE-Net simultaneously satisfies the stringent clinical requirements of low false-alarm rates (FP) and low missed-detection rates (FN) in real-world monitoring scenarios.

6.2.3. Comparison between our proposed LAVAE-Net model and other models

Table 10 compares the accuracy of LAVAE-Net with that of existing methods. Pereira and Silveira [30] trained their VRAE-based models on both normal and abnormal beats (400/100/4,500 split) and reached 98.43% with VRAE + SVM, whereas their VRAE + Wasserstein and VRAE + Clustering variants achieved 95.96%. Murat et al. [32] employed a CNN-LSTM trained on 4,000 normal and abnormal samples and obtained 93.90%. Roy et al. [22], in contrast, trained only on normal data (2,101/234/2,665 split) and reported 98.68% with a manual threshold and 98.42% with an automatic Kapur threshold. Using the identical normal-only split, our LAVAE-Net attains 99.06%, thereby exceeding all previous results.

6.2.4. Reconstruction analysis

Fig. 13 presents two reconstruction plots: a normal ECG segment on the left and an abnormal segment on the right. In each subplot, the original signal (blue) is overlaid with the model-generated reconstruction (orange), illustrating the model's ability to reproduce both types of waveforms. For the normal segment, the orange curve closely follows the blue input, yielding a reconstruction loss of only 42.77. This low error confirms that the model has effectively captured the overall

morphology and key features of normal heartbeats. Conversely, the abnormal segment shows a pronounced mismatch between the original and reconstructed curves, reflected in a much higher loss of 160.05. The large discrepancy demonstrates that the model struggles to reproduce patterns it has never seen during training.

The stark contrast arises because the network was trained solely on normal ECG samples. To narrow the gap on abnormal data, future work could refine the architecture or incorporate additional arrhythmic examples, thereby enhancing the model's adaptability and accuracy.

6.3. Threshold selection

In clinical ECG monitoring, the risk of missed detection of cardiac abnormalities outweighs the consequences of false alarms, as false negatives may directly endanger patients' lives, whereas false positives primarily increase healthcare costs and operational burdens. To balance the recall and precision of anomaly detection, the threshold is usually set at the point that maximizes the F1 score of the anomalous class, which is generally located in the intersection area of the anomalous error distribution and the normal error distribution, and is shifted towards the direction of "reducing missed detections" to lower the false negative rate. This strategy makes the model more prone to misclassifying borderline normal samples as abnormal, deliberately trading a modest reduction in specificity for substantially enhanced sensitivity—a clinically necessary compromise.

In practical applications, arrhythmia discrimination is further refined by incorporating clinical prior knowledge to dynamically adjust the threshold according to the clinical severity of different types of arrhythmias. For instance, for life-threatening arrhythmias such as ventricular tachycardia, the threshold is intentionally lowered to maximize sensitivity; for arrhythmias of minor clinical significance, the threshold is raised to reduce false-positive alarms. This strategy allows the model to align closely with real clinical needs and enhances the clinical applicability of the ECG monitoring system.

6.4. Selection of loss function

We used the Huber + KL loss function and other loss functions, and the performance comparison is shown below:

In Table 11, the model trained with the MSE loss function performs worse, which can be attributed to the high amplitude and short duration of the QRS complex in ECG signals. Traditional MSE yields excessively large gradients for these peak anomalies, hampering model convergence [29]. By contrast, the Huber loss behaves quadratically for small errors and linearly for large ones, ensuring smooth gradients near the optimum while attenuating the undue influence of outliers; consequently, it achieves superior performance.

6.5. Limitations and future work

In this paper, the 95th percentile of training loss values is used as a threshold during training to distinguish between normal and abnormal samples, which may fail to account for actual model performance variations. For certain datasets or anomaly types, such a threshold might be insufficient to capture all anomalous samples, leading to reduced accuracy. Because the threshold is derived only from normal-sample reconstruction errors, any anomalous signal that appears can skew the error distribution and blur the separation between normal and abnormal instances. Moreover, the current model flags anomalies without distinguishing their types. In future work, semi-supervised learning could be adopted to refine the threshold and enable fine-grained anomaly classification.

7. Conclusion and future work

In this study, we successfully developed and validated an effective

Table 10
Performance comparison in terms of accuracy of LAVAE-Net and other models.

Method	Model	Splitting	Accuracy
Pereira and Silveira[27]	VRAE + SVM	#training normal & anomaly ECGs = 400 #validation normal & anomaly ECGs = 100 #test normal & anomaly ECGs = 4500	98.43%
	VRAE + Wasserstein	#training normal & anomaly ECGs = 400 #validation normal & anomaly ECGs = 100 #test normal & anomaly ECGs = 4500	95.96%
	VRAE + Clustering	#training normal & anomaly ECGs = 400 #validation normal & anomaly ECGs = 100 #test normal & anomaly ECGs = 4500	95.96%
Murat et al.[28]	CNN-LSTM	#training normal and anomaly ECGs = 4000 #validation Normal & anomaly ECGs = 500 #test normal & anomaly = 500	93.90%
Roy et al.[22]	ECG-NET with manual threshold	# training normal ECGs only = 2101 #validation normal ECGs only = 234 # test normal & anomaly = 2665	98.68%
	ECG-NET with automated threshold	#training normal ECGs only = 2101 #validation normal ECGs only = 234 # test normal & anomaly = 2665	98.42%
Proposed LAVAE-Net	LSTM-Attention VAE	#training normal ECGs only = 2101 #validation normal ECGs only = 234 # test normal & anomaly = 2665	99.06%

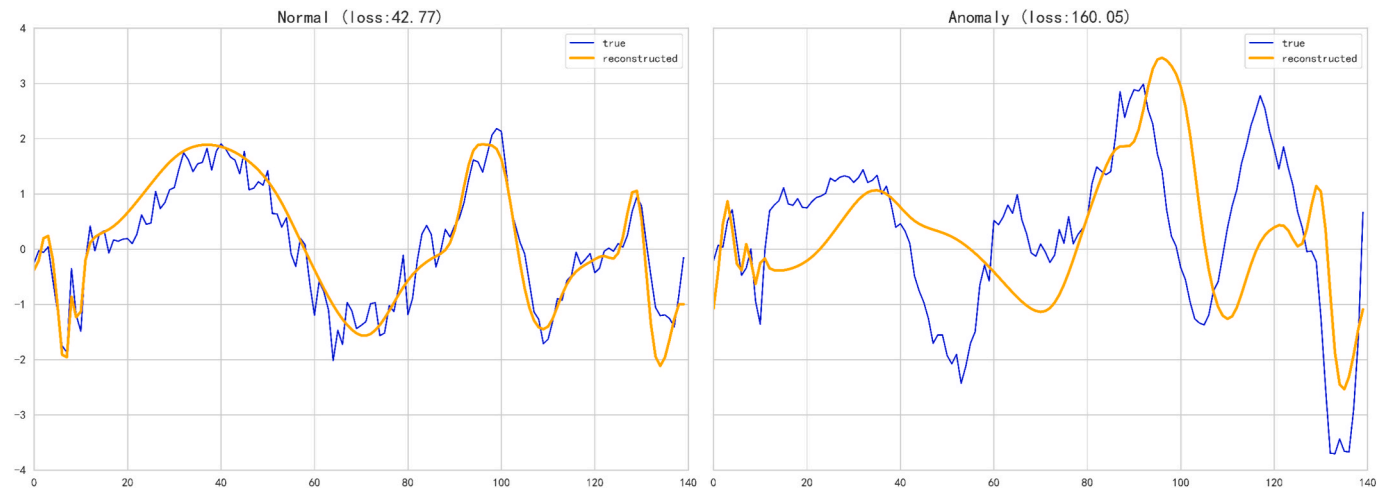


Fig. 13. Reconstruction curves of normal and abnormal ECG.

Table 11
F1_{Anomaly} score and F1_{Anomaly} score for different loss functions.

	MSE	MSE + KL	Huber + KL
F1 _{Anomaly}	82.36%	91.36%	99.40%

ECG anomaly detection method to address the deficiency of a small amount of abnormal ECG data in the clinical environment. This method is based on an attention mechanism-enhanced VAE structure, which is trained only with normal ECG data to learn normal ECG features and identifies abnormal signals through reconstruction error. The experimental results fully demonstrate the effectiveness of the proposed method. On the ECG5000 dataset, our model achieved F1-Anomaly of 99.40%, with normal/abnormal detection accuracies of 97.35% and 96.21% respectively. From the perspective of clinical application, the proposed method has important practical significance. The model does not need to collect and label a large number of abnormal ECG samples in advance, which greatly reduces the cost and complexity of system deployment. More importantly, because the model identifies abnormalities by learning normal patterns, it has the potential to detect novel or rare cardiac abnormalities that have not appeared in the training data, which is of great value for discovering unknown disease patterns in clinical diagnosis.

Looking ahead, this study opens up new research directions in the field of ECG anomaly detection. Validating the model's generalization capabilities on larger-scale and more diverse ECG datasets will be a critical next step. Simultaneously, extending this approach to multi-lead ECG analysis and exploring the interpretability of attention mechanisms will further enhance the model's clinical value. Moreover, applying this technology to real-time ECG monitoring systems to enable continuous cardiac health surveillance represents a promising application direction. It holds significant potential to play a critical role in future cardiac disease diagnosis and health monitoring systems.

In future work, we plan to integrate IGANomaly, FGANomaly, and BeatGAN[30] with LAVAE-Net and conduct a comprehensive evaluation to improve the overall performance of ECG anomaly detection. These advanced models, which leverage generative adversarial networks and frequency-domain representations, provide strong baselines for comparison due to their ability to model rare and subtle abnormal patterns in ECG signals. By combining these state-of-the-art techniques, we aim to enhance the robustness and accuracy of our detection system, paving the way for even more efficient and accurate clinical applications. Additionally, we will evaluate the system using the MIT-BIH database as well as other relevant datasets, allowing for a broader and more diverse analysis of the model's performance across different populations and

clinical scenarios.

Code and data.

Our code, data, and model checkpoints are available here: <https://github.com/cropsoft-shi/ECGTimeSeries>.

CRedit authorship contribution statement

Hao Shi: Writing – original draft, Methodology. **Zhuoying Jiang:** Writing – review & editing, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link our data and code on github.

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